CAREFLOW OPD

FINAL PROJECT REPORT

The Hospital Booking System for the Outpatient Department

6931503085 Apichaya Somrit

6931503092 Aung Phone Thant

6931503093 Min Hein Ko

6931503098 Yoon Yadanar MyintHtay

6931503116 Varichsara Kitiphatphiboon

Repository: https://github.com/Min-Hein-Ko/The-hospital-booking-system

Live Demo: https://min-hein-ko.github.io/The-hospital-booking-system/

Academic front-end prototype using sample data. Not intended for real patient data or clinical operation.

01 | PROJECT CONTEXT

Problem & Users

CareFlow OPD is a browser-based prototype designed to make outpatient appointment information and booking tasks easier to access. It focuses on the patient journey from finding a doctor to managing an appointment and checking a personal queue token.

Problem & Users

Patients need a convenient way to:

- Find doctors and check availability.
- Book, cancel or reschedule appointments.
- Receive a token and check their own appointment status.
- Manage their profile and notifications.

Admins need a clear way to:

- Maintain doctor information and schedules.
- Monitor appointments, tokens and dashboard summaries.
- Manage patient account access.
- Review records without manually controlling the queue.

Project Scope

In scope: patient registration/sign-in/profile, doctor discovery, appointment booking and management, personal token/status display, doctor and schedule administration, read-only Admin monitoring, patient account Ban/Restore, and in-app notifications.

Out of scope: billing, telemedicine, diagnosis/treatment advice, prescriptions, full electronic medical records, emergency services, insurance claims, online pharmacy and external hospital integration.

The defined system roles are Patient and Admin. Doctor is a profile/stakeholder record.

Functional Requirements & Proposed MoSCoW

ID	Requirement summary	Priority
FR-1	Patients view doctor information and availability.	Must
FR-2	Patients book appointments, provide optional visit reasons, and receive tokens.	Must
FR-3	Patients view their own token numbers and appointment statuses.	Must
FR-4	Patients cancel, reschedule, or arrange follow-up appointments.	Must
FR-5	Admins manage doctor information and schedules.	Must
FR-6	Admins monitor dashboards, appointments, and token records.	Must
FR-7	Admins ban or restore patient accounts.	Must
FR-8	Patients register, log in, and manage their profiles.	Must
FR-9	Patients and Admins receive and manage notifications	Must

Measurable Non-Functional Requirements

ID / Type	Requirement	Verification
NFR-1 Performance	Doctor and appointment schedule pages shall load within 3 seconds under normal network conditions.	Measure page load time under normal network conditions.
NFR-2 Performance	The system shall create and display a patient's token number within 2 seconds after booking confirmation.	Measure the time between booking confirmation and token display.
NFR-3 Usability	A first-time user shall be able to find a doctor and complete an appointment booking in under 2 minutes without assistance.	Conduct a timed trial with a first-time user.
NFR-4 Reliability	The system shall save a confirmed appointment and its token number so that the records remain available after the user leaves or refreshes the appointment page.	Create an appointment, refresh the page, and verify the appointment and token records.
NFR-5 Security / Privacy	A user shall only be able to view and manage their own personal information and booking history.	Log in as one user and attempt to access another user's information; access must be prevented.

02 | SYSTEM DESIGN

Architecture & Design

The prototype is a front-end demonstration built with HTML5, CSS3 and JavaScript and published through GitHub Pages. It uses sample records in browser storage. The logical data model describes the target information structure.

Logical Architecture

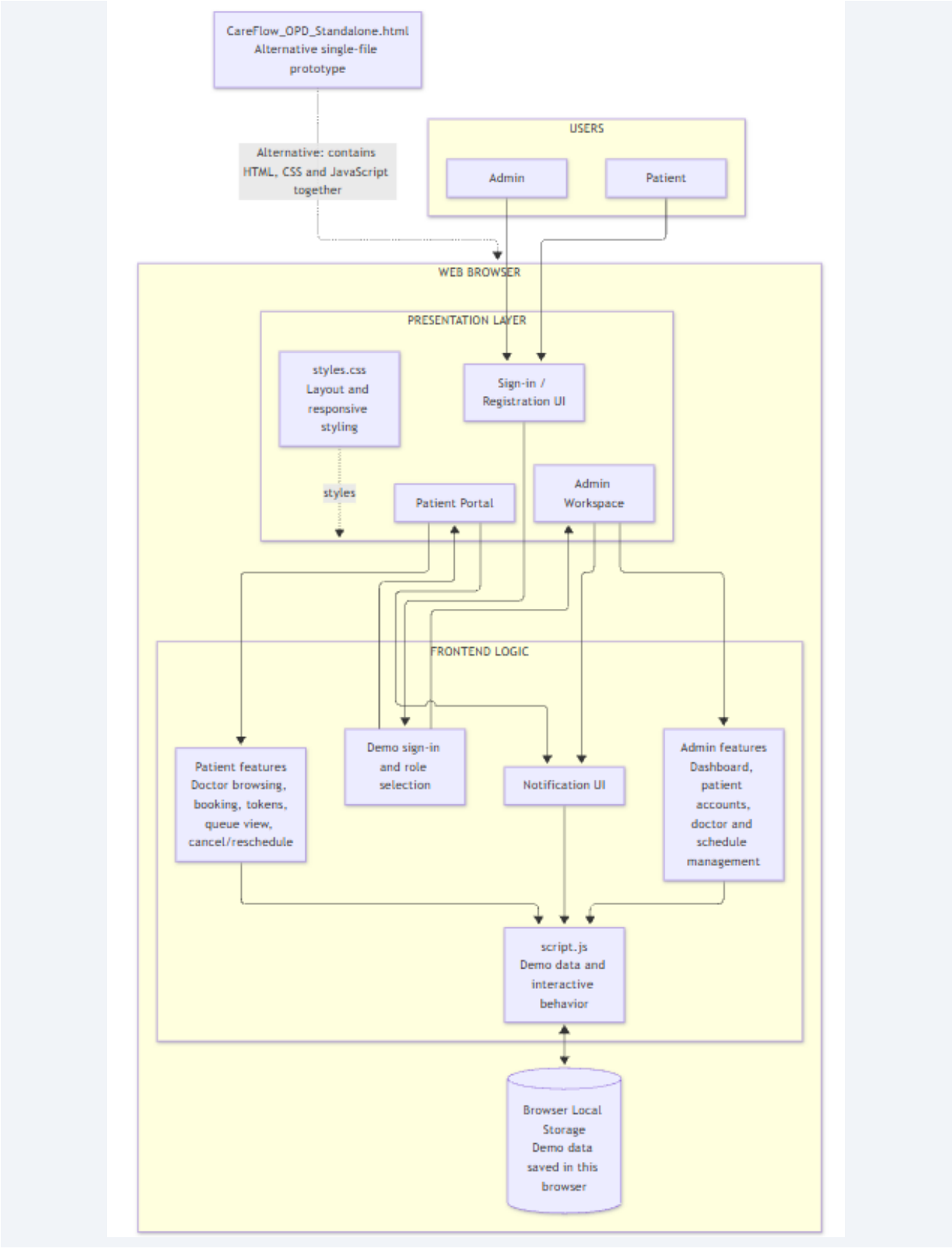


Figure 1. CareFlow OPD logical architecture.

Use Case Diagram

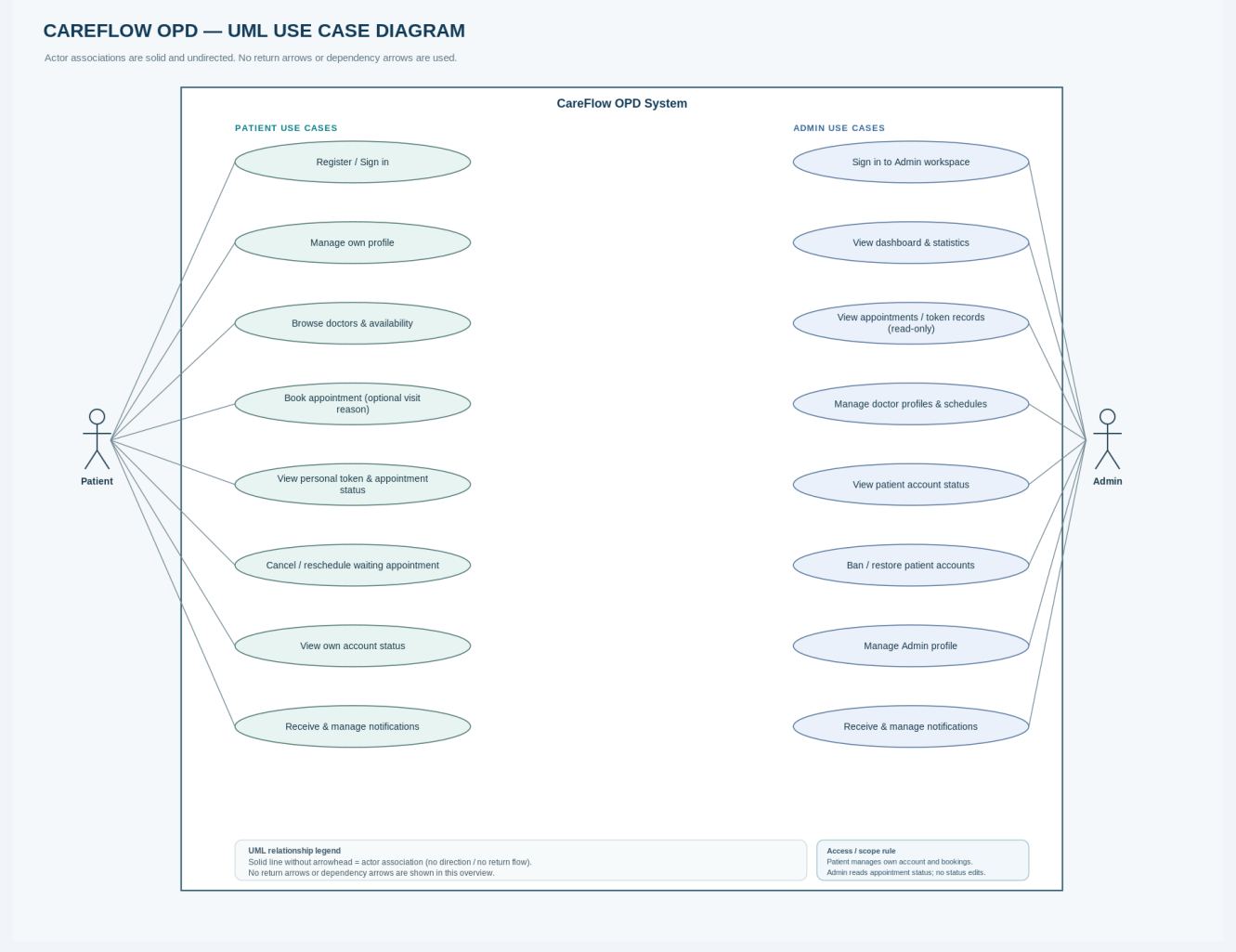


Figure 2. CareFlow OPD use-case diagram.

Sequence Diagram

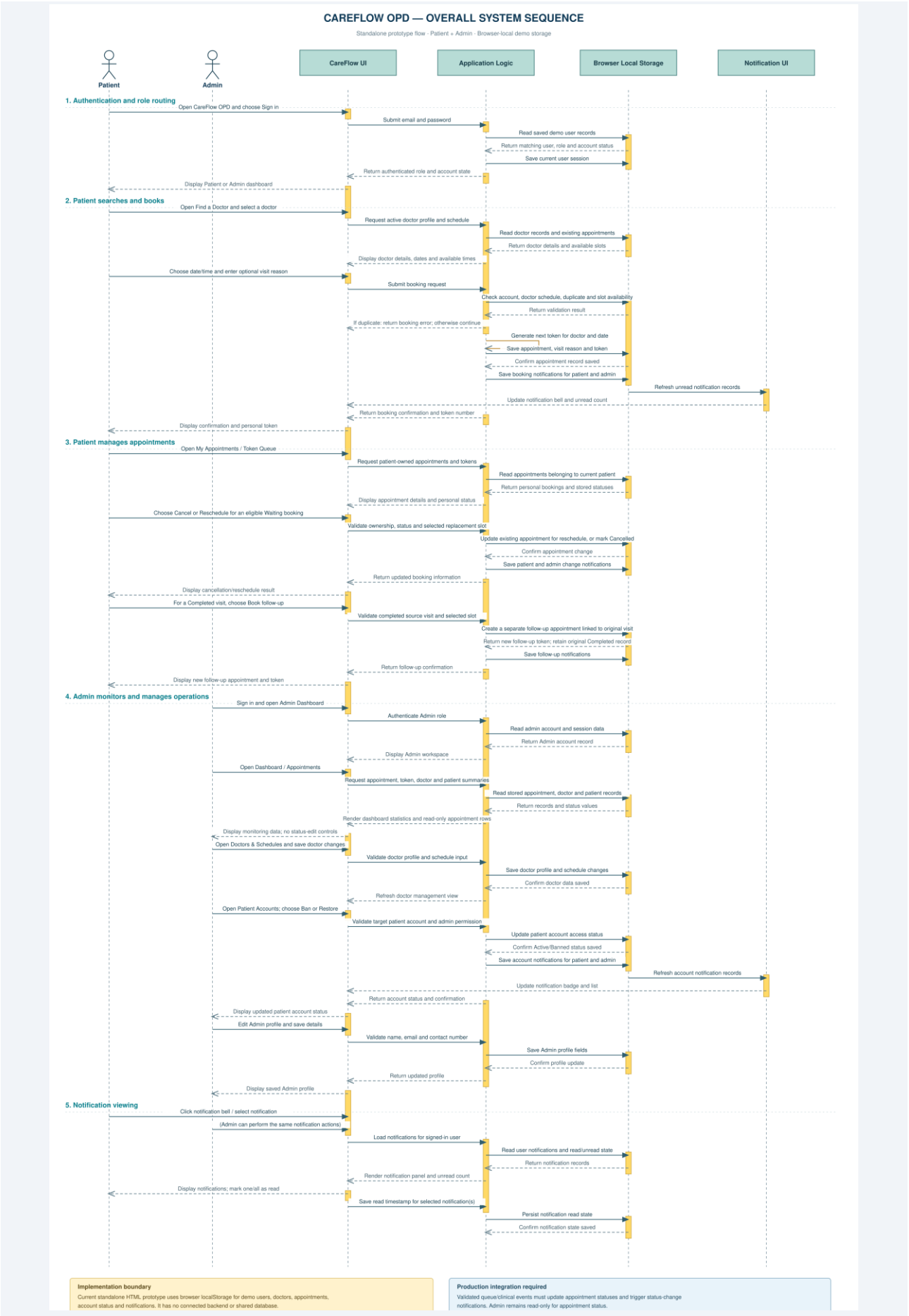


Figure 3. Sequence diagram.

03 | DATA MODEL AND PROTOTYPE

Data Model & Prototype Build

Logical Data Model

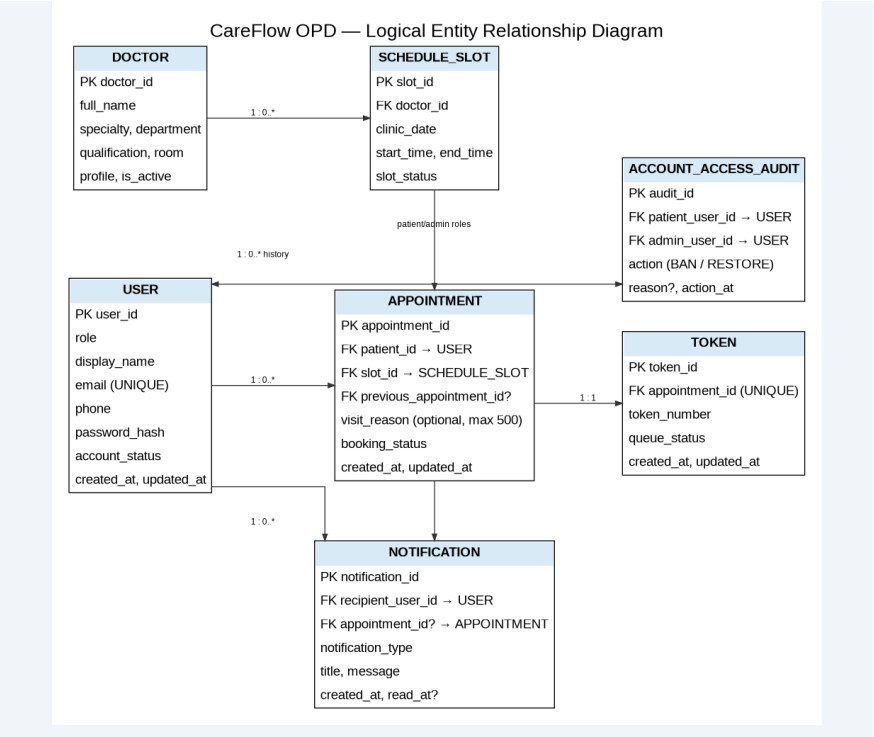


Figure 4. CareFlow OPD logical entity-relationship diagram.

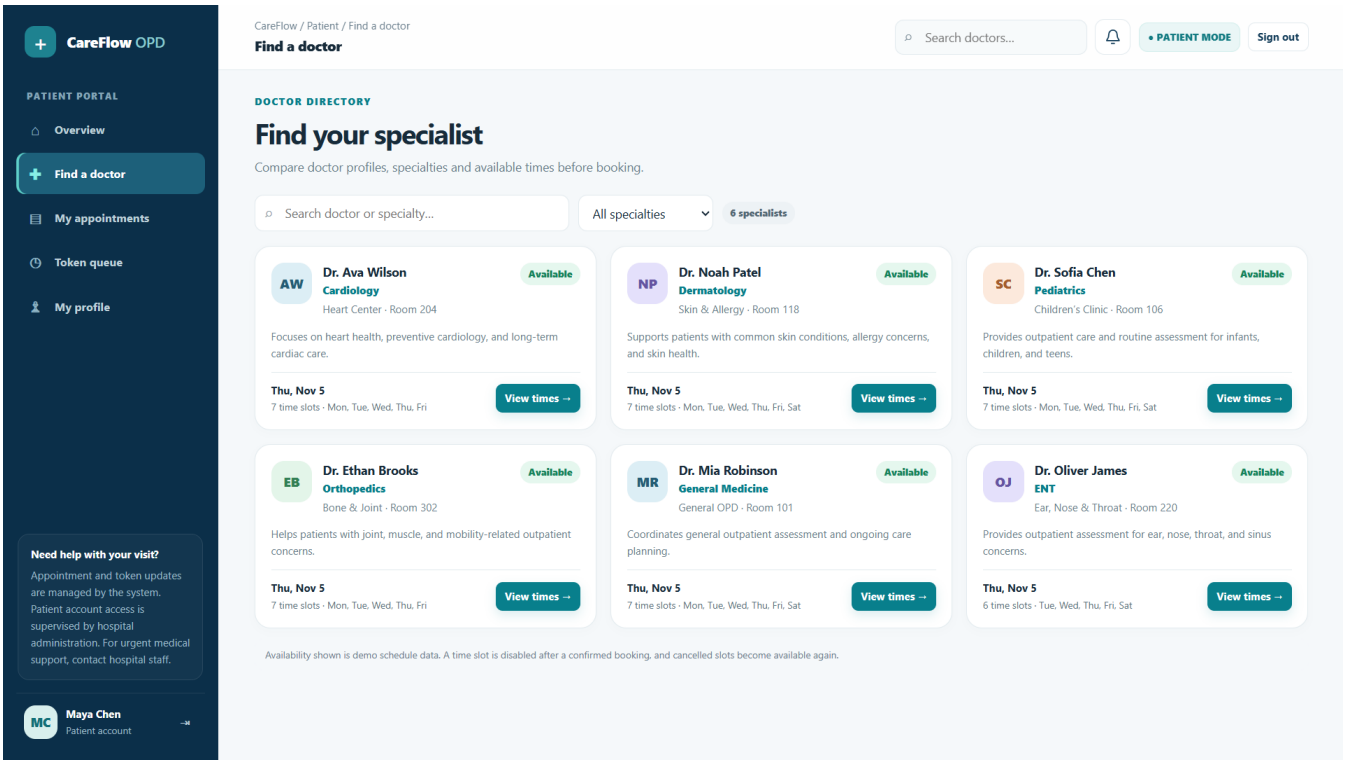
Prototype Build

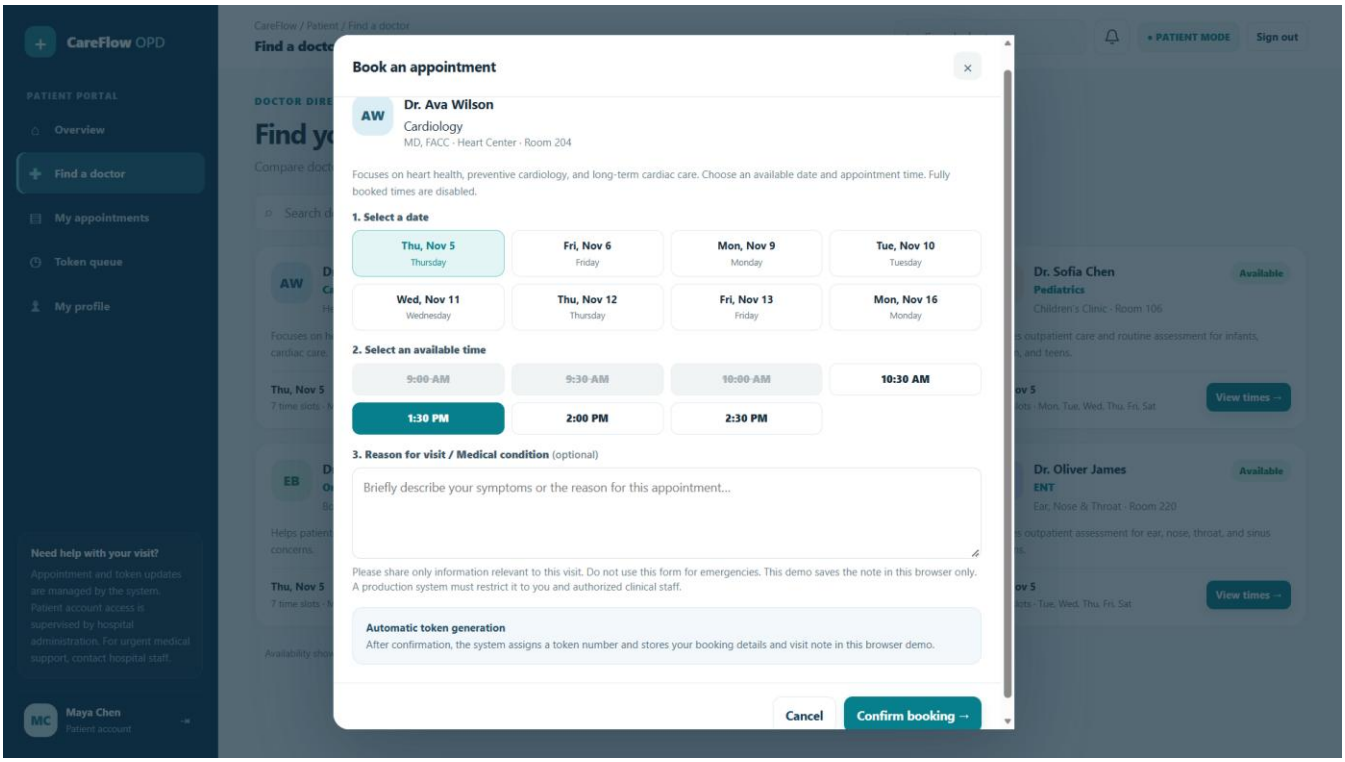
Item	Project information
Front-end technologies	HTML5, CSS3 and JavaScript
Repository / hosting	GitHub repository; published demo via GitHub Pages
Data handling	Sample records stored in browser for demonstration
User roles	Patient and Admin
Prototype boundary	Front-end demonstration only; not a production hospital system or completed backend

The prototype demonstrates intended screens and interactions. It does not prove production-grade shared persistence, backend authentication/authorization, atomic concurrent booking, durable audit logging or external notification delivery.

Prototype Screenshots — Insert Actual Browser Captures

Prototype screenshots (Patient)





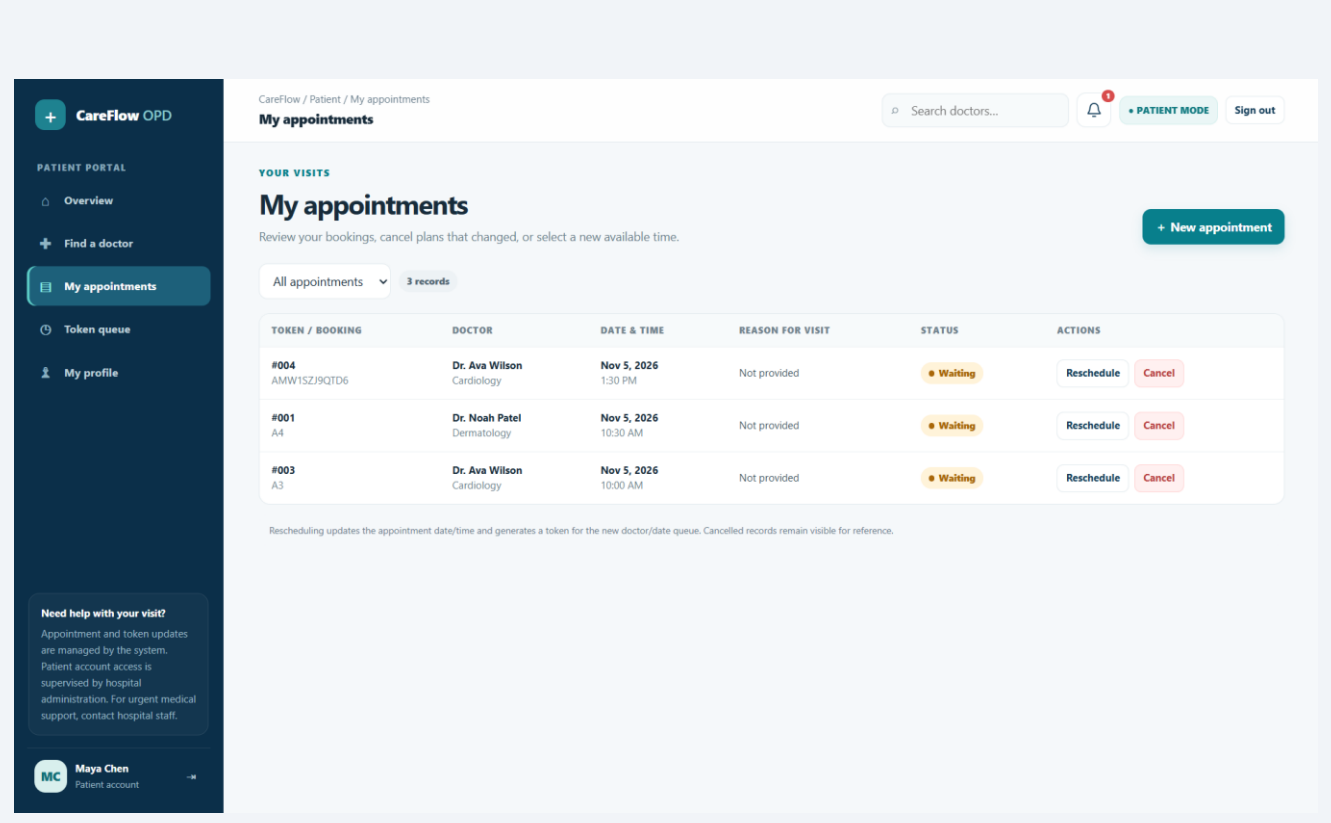
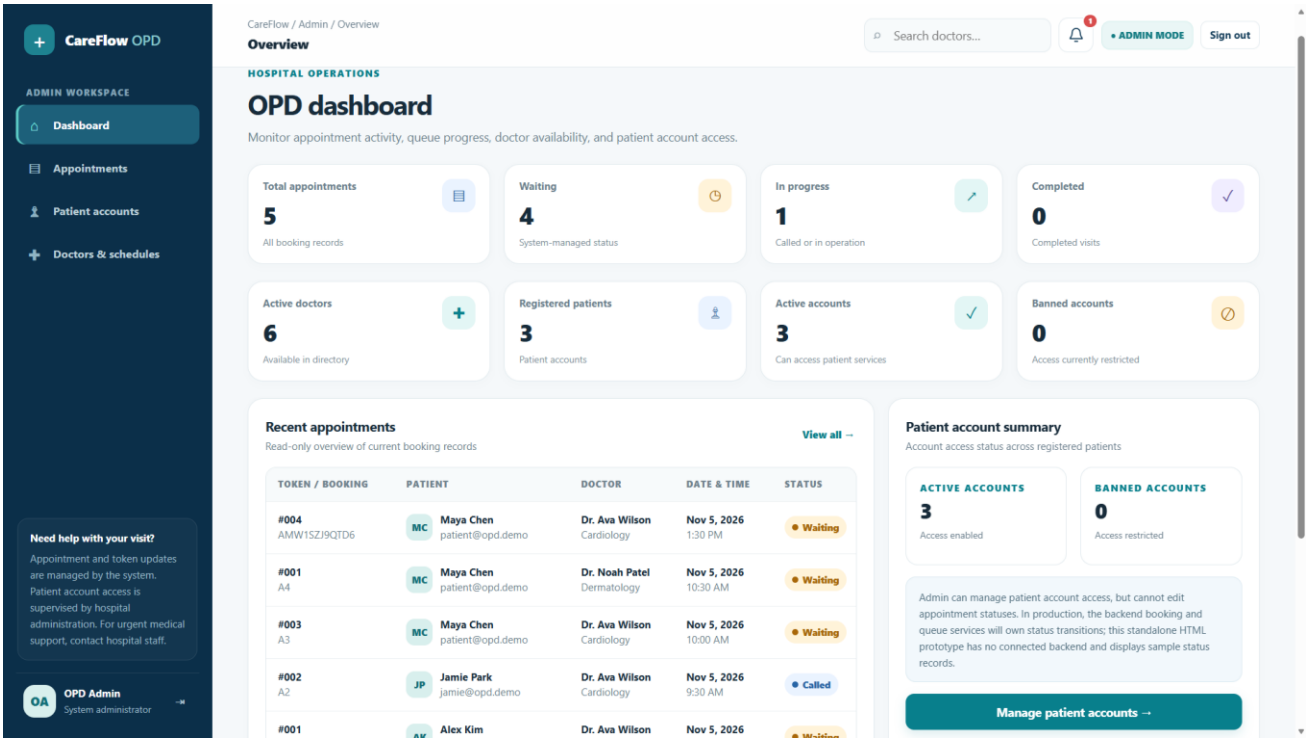


Figure 5-7. Prototype screenshots (Patient)

Prototype screenshots (Admin)



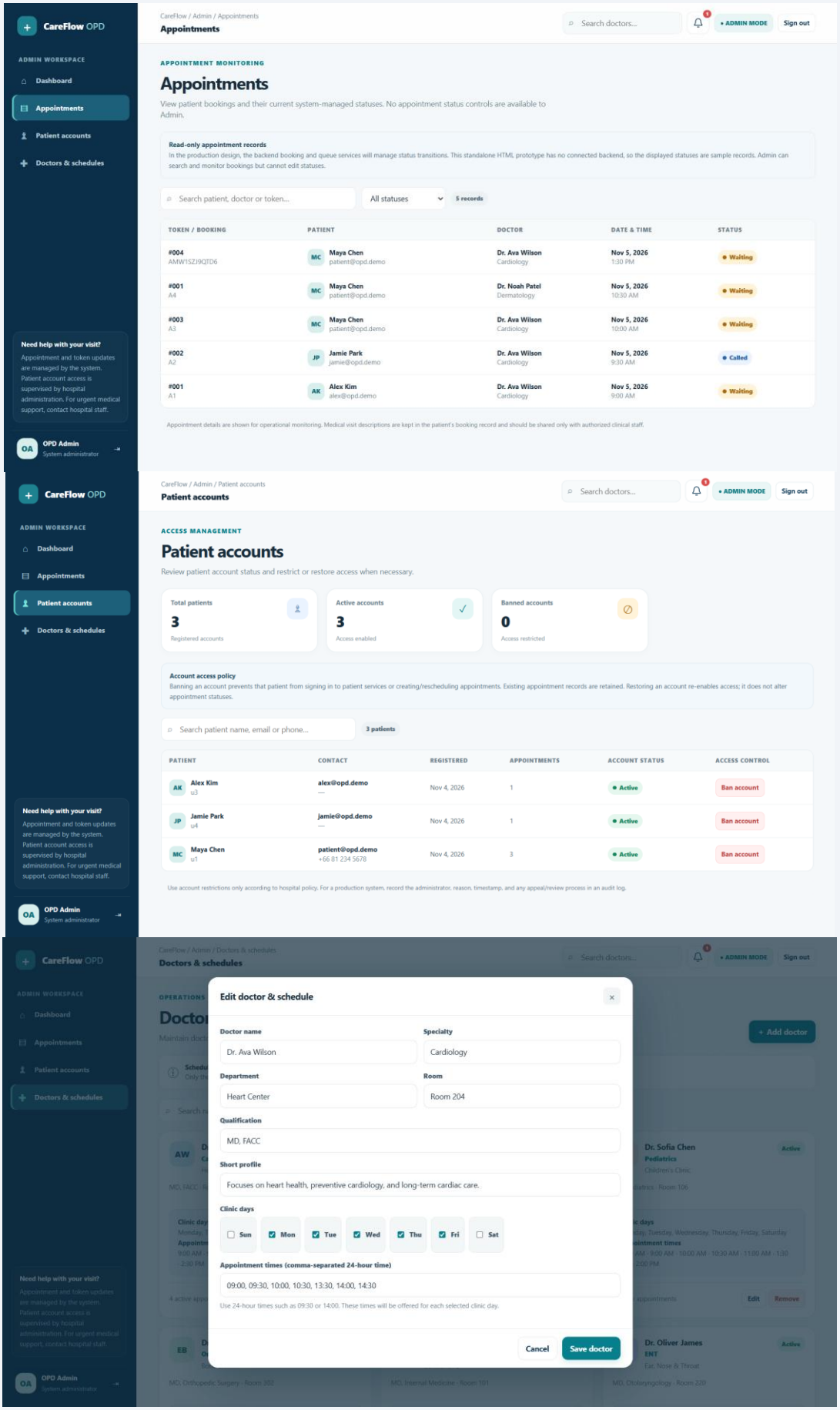


Figure 8-10. Prototype screenshots (Admin)

04 | VERIFICATION

Testing Results — FR-1 to FR-5

Results below are organized from the team's supplied testing notes. “Pass (reported)” means the team reported the criterion as working in the prototype; it is not an independent re-test by the report author.

FR	Test scenario	Expected result	Actual result
FR-1	Open the doctor directory using the patient interface.	The directory presents doctors who are active and identifies each doctor's specialty.	Pass. Active doctors and their specialties can be viewed
FR-1	Open a doctor's booking options and inspect the available dates and times.	The patient can see the appointment dates and time slots offered for booking	Pass. Available dates and times are displayed to the patient
FR-1	Attempt to choose a doctor or time slot that is already unavailable because it has been booked	The unavailable doctor or time slot is disabled or otherwise prevented from being selected.	Pass. A time slot that has already been booked cannot be selected.
FR-2	Complete the booking selection using an available doctor, date, and time	The system accepts the selection when all chosen options are available.	Pass. A patient can select an available doctor, date, and time.
FR-2	Submit an appointment booking and inspect the confirmation.	The system creates the appointment and assigns a token that begins in the Waiting state. It should also prevent duplicate bookings.	Partially verified. A token is generated after a successful booking. The recorded observation does not separately demonstrate the initial status or duplicate-booking prevention.
FR-3	View the details shown after making an appointment.	The appointment view identifies the patient's token number, selected doctor, appointment date, and appointment time.	Pass. The token number and appointment date and time are shown after booking; the doctor is selected as part of the displayed appointment information
FR-3	Check the appointment after its status has been changed by the system.	The patient-facing view reflects the appointment status supplied by the system.	Pass. The prototype displays statuses such as Waiting and Cancelled in the observed situations
FR-3	Sign in as a patient and inspect the token information available in the account.	The patient can access their own appointment and token information, but not another patient's private token details.	Pass. The tested patient account did not provide access to other patients' token information
FR-4	Cancel an appointment that is still upcoming.	The appointment is cancelled and its status changes to Cancelled.	Pass. The appointment status changes to Cancelled after cancellation
FR-4	Reschedule an existing appointment by choosing another available date and time.	The system saves the new appointment time and issues a new token for the rescheduled booking.	Pass. A new available date and time can be selected, and a new token number is generated.
FR-4	Create a follow-up appointment after an earlier appointment has been completed	The new booking is stored as a separate appointment, while the completed appointment remains in the patient's history.	Not tested. The team did not perform this scenario because the observed appointment flow ended with a Completed status and no follow-up booking was required.
FR-5	Use the administrator interface to edit a doctor's details and schedule.	An administrator can update the doctor's information and availability.	Pass. Doctor information and schedules can be edited from the administrator interface
FR-5	After an administrator saves a doctor or schedule change, view the same information as a patient	The patient interface displays the updated doctor information and availability.	Pass. The changes saved by the administrator are visible from the patient interface.
FR-5	Attempt to remove a doctor who has an existing active appointment.	The system blocks deletion when removing the doctor would affect an existing appointment record	Pass. A doctor with an existing active appointment cannot be removed.

04 | VERIFICATION (CONTINUED)

Testing Results — FR-6 to FR-9

FR	Acceptance criterion	Result	Team observation
FR-6	Open the administrator dashboard and review its summary information.	The dashboard provides an overview of appointment activity, active doctors, and patient accounts	Pass. The administrator dashboard can be used to monitor the system overview.
FR-6	Open the appointment and token information from the administrator interface.	The administrator can inspect appointment and token information, including locating relevant records where the interface provides search or filtering	Partially verified. The team confirmed that the administrator can check the records. The recorded observation does not separately confirm every search and filter operation.
FR-6	Inspect appointment status and token information from the administrator dashboard.	The administrator can view this information but cannot manually change appointment statuses or control the token queue through the dashboard.	Pass. Appointment and token information are view-only in the administrator dashboard.
FR-7	Open a patient's account details from the administrator interface.	The administrator can view patient information and identify whether the account is Active or Banned.	Pass. The administrator can check an individual patient's account status.
FR-7	Ban a patient account and then use the restore option.	The administrator can restrict the account and later restore its access	Partially verified. The ban and Restore Access actions were successfully observed. The recording does not separately confirm an audit log of these actions.
FR-7	Try to use appointment services while signed in to a banned patient account.	The system blocks the account from booking or rescheduling and displays an access-restriction message.	Pass. The banned account displayed an account-access restriction message
FR-8	Check the entry points for a new and an existing patient.	A new patient can start registration, while a registered patient can access the sign-in option	Pass. The interface provides Create an Account and Sign In options.
FR-8	Open the patient profile and edit personal details.	The patient can update their own profile information, including name, email address, and phone number.	Pass. The tested profile allowed these personal details to be edited.
FR-8	Attempt to access booking or rescheduling while the patient account is banned.	Restricted accounts cannot use appointment booking or rescheduling until access is restored.	Pass. The banned account was prevented from using the system and showed an access-restriction message.
FR-9	Open the notification area and inspect the unread notification indicator	The interface displays notifications and indicates how many are unread.	Pass. Notifications can be opened, and an unread count is displayed
FR-9	Open a notification and use the control to mark notifications as read	The user can read notifications and mark an individual notification or all notifications as read	Pass. The team confirmed that the Mark All as Read control can be used to clear the unread state.
FR-9	Reschedule an appointment and inspect the notification received by the patient.	The system sends a relevant notification after the change and does not expose the patient's visit-reason description in the notification	Partially verified. A notification related to a successful reschedule was observed. Notifications for every listed event type and the exclusion of visit-reason information were not separately documented as tested.

Test outcome summary

Outcome	Number of tests	Meaning
Pass	21	The recorded observation supports the expected behavior for the tested scenario.
Partially verified	4	Some behavior was observed, but one or more parts of the combined expected result were not separately confirmed in the test record.
Not tested	1	The team did not execute the scenario.
Total	**26**	**All test scenarios listed in this document.**

05 | RESPONSIBLE AI USE

AI Usage Log

AI was used as a support tool for brainstorming, organization, explanation, drafting and troubleshooting. The team retained responsibility for project decisions, reviewing outputs, checking consistency and deciding what to include.

AI Usage Record: M1–M3

Deliverable	How ChatGPT supported the work	Team contribution and review
M1 — Project idea and initial definition	Brainstormed and compared project directions; suggested ways to clarify the OPD problem, purpose, scope and document structure.	The team selected the hospital OPD booking topic, supplied context, decided scope, revised wording and checked relevance and consistency.
M2 — Requirements and analysis	Helped organize requirement sections, improve wording and explain FR/NFR, user stories, acceptance criteria, roles, booking and token-queue features.	The team determined requirements, adjusted them to the agreed workflow, and reviewed clarity, feasibility, duplication and scope consistency.
M3 — Design and project refinement	Provided writing/review support, design explanations, section-organization suggestions and feedback on consistency between requirements and proposed design.	The team supplied project decisions, incorporated relevant suggestions and reviewed M3 against system features and earlier documentation. ChatGPT was not the authority for technical or operational decisions.

AI Usage Record: Diagrams, Prototype and Presentation

Work product	How ChatGPT supported the work	Team review before repository push
UML and system diagrams	Explained UML conventions; suggested improvements to actors, use-case relationships, appointment/token flows, sequence order, include and extend relationships.	The team checked actors, boundaries, labels, flow direction and UML relationships against requirements; confusing elements were revised.
Front-end prototype	Supported HTML, CSS and JavaScript through code explanations, layout/wording suggestions and troubleshooting guidance.	The team opened the prototype locally and checked rendering, navigation, interactions, labels and patient/Admin requirements. Changed files were reviewed before commits and pushes.
Demo / presentation slides	Helped organize presentation flow, summarize project information and improve slide wording and clarity.	The team compared slides with M1–M3, diagrams and prototype, and checked that claims did not exceed documented scope.
GitHub repository preparation	Provided guidance on file organization, folder structure and general repository practices when requested.	The team reviewed files, names and folder placement, opened key documents/diagrams/prototype files, corrected visible issues and pushed after team review.

AI Use and Accountability

AI-generated suggestions were reviewed rather than accepted automatically. The team remained responsible for technical decisions, prototype behaviour, source accuracy, repository content and final submission. AI assistance does not establish that a feature was implemented or tested; implementation and test claims are limited to team-supplied observations and available project evidence.

06 | OWNERSHIP AND TRACEABILITY

Team Contributions & Golden Thread

Individual Contributions

Team member	Github	Contribution
Min Hein Ko	Min-Hein-Ko	M1, M2, Prototype, Diagrams, README
Yoon Yadanar Myint Htay	yoonymh1	M1, M2, Specification, UI
Varichsara Kitiphatphiboon	jazzvarichsara-stack	M1, M2, UI, Process and Team Coordination
Aung Phone Thant	6931503092-lgtm	M1, M2, Functional Requirements (FR), Non-Functional Requirements (NFR), Use Cases
Apichaya Somrit	inriSSh	M1, M2, Data Model, Screen List, Traceability Summary (Golden Thread) in the SRS

Golden Thread Traceability

Problem / Need	Requirement	design	Feature	Test
Patients need to find doctors and check availability.	FR-1	SC-04 Find a Doctor	Doctor search, specialty filter, active doctor list, and available appointment slots.	**Pass**: Active doctors, specialties, available dates and times were displayed. Unavailable doctors and booked time slots could not be selected.
Patients need to book appointments and receive a token.	FR-2, FR-3	SC-04 Find a Doctor → SC-06 Token Queue	Book an appointment, provide an optional visit reason, generate a token, and display booking confirmation.	**Partial Pass**: Patients could select available booking options and received a token after successful booking. Token uniqueness and duplicate-booking prevention were not fully verified.
Patients need to know their queue progress.	FR-3	SC-06 Token Queue	Display the patient's own token number and current queue status.	**Pass**: The patient's own token and appointment details were displayed. Other patients' tokens could not be viewed. Waiting and Cancelled statuses were observed.
Patients may need to change appointments or book a follow-up.	FR-2, FR-4	SC-05	SC-05 My Appointments	**Partial Pass**: Cancellation updated the status to Cancelled. Rescheduling to another available slot generated a new token. Follow-up booking was **Not tested** because no follow-up was needed during testing.
Admins need to maintain accurate doctor information and schedules.	FR-5	SC-11 Doctors & Schedules	Add, edit, and remove (deactivate) doctor records; manage clinic days and appointment times.	**Pass**: Admins could edit doctor information and schedules, and the updated information appeared in the patient view. Permanent removal of doctors with active appointments was prevented.
Admins need to monitor appointments and patient accounts without changing appointment or queue statuses.	FR-6, FR-7	SC-08 Admin Dashboard → SC-09 Admin Appointments → SC-10 Patient Accounts	View dashboard statistics and appointment records; Ban or Restore patient accounts.	**Pass**: Admins could monitor dashboard information and view appointment/token records without editing them. Admins could view account status, Ban/Restore accounts, and banned patients were blocked from access.
Patients need to create accounts and manage their personal information.	FR-8	SC-01 Sign In / Register → SC-07 My Profile	Register, log in, and update profile information.	**Pass**: Sign-in and account-creation options were available. Patients could edit their name, email and phone number. Banned accounts were restricted.
Patients and Admins need to receive and manage important system updates.	FR-9	SC-03 Patient Overview / SC-08 Admin Dashboard → Notification Panel	Display notifications, unread count, notification list, and Mark as Read actions.	**Partial Pass**: Users could view notifications and unread counts, and mark all as read. A notification after rescheduling was observed. Other notification types and the exclusion of visit-reason information still need verification.